CSHLvc2026

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Welcome

Package: CSHLvc2026 **Authors:** Vincent Carey [aut, cre] **Compiled:** 2026-06-12 **Package version:** 0.98.2 **R version:** R version 4.6.0 (2026-04-24) **BioC version:** 3.24 **License:** MIT + file LICENSE

This is the landing page of the `BiocBook` entitled ….

This book introduces the reader to ….

Docker image

A `Docker` image built from this repository is available here:

ghcr.io/vjcitn/cshlvc2026

Get started now

You can get access to all the packages used in this book in $<$ 1 minute, using this command in a terminal:

Listing 0.1 `bash`

```bash
docker run -it ghcr.io/vjcitn/cshlvc2026:devel R
```

RStudio Server

An RStudio Server instance can be initiated from the `Docker` image as follows:

Listing 0.2 `bash`

```
docker run \
    --volume <local_folder>:<destination_folder> \
    -e PASSWORD=OHCA \
    -p 8787:8787 \
    ghcr.io/vjcitn/cshlvc2026:devel
```

The initiated RStudio Server instance will be available at https://localhost:8787.

Session info

Click to expand

1 Introduction to GWAS

This chapter introduces genome-wide association studies (GWAS), covering the conceptual foundations and practical tools needed to design, execute, and interpret large-scale genetic association analyses.

1.1 Phenotype Vocabulary

At the boundaries of knowledge, it is important to establish conventions to support clear communication. “Ontology” is a field of information science devoted to formal specification of vocabularies and associated conceptual hierarchies. Practical use of ontologies in genomic data science started with Gene Ontology but has grown to involve frequent use of ontologies for

- cell types
- diseases
- experimental factors and designs

Here is an “ontological report” on “exocrine pancreatic carcinoma”, provided by the European Bioinformatics Institute Ontology Lookup Service:

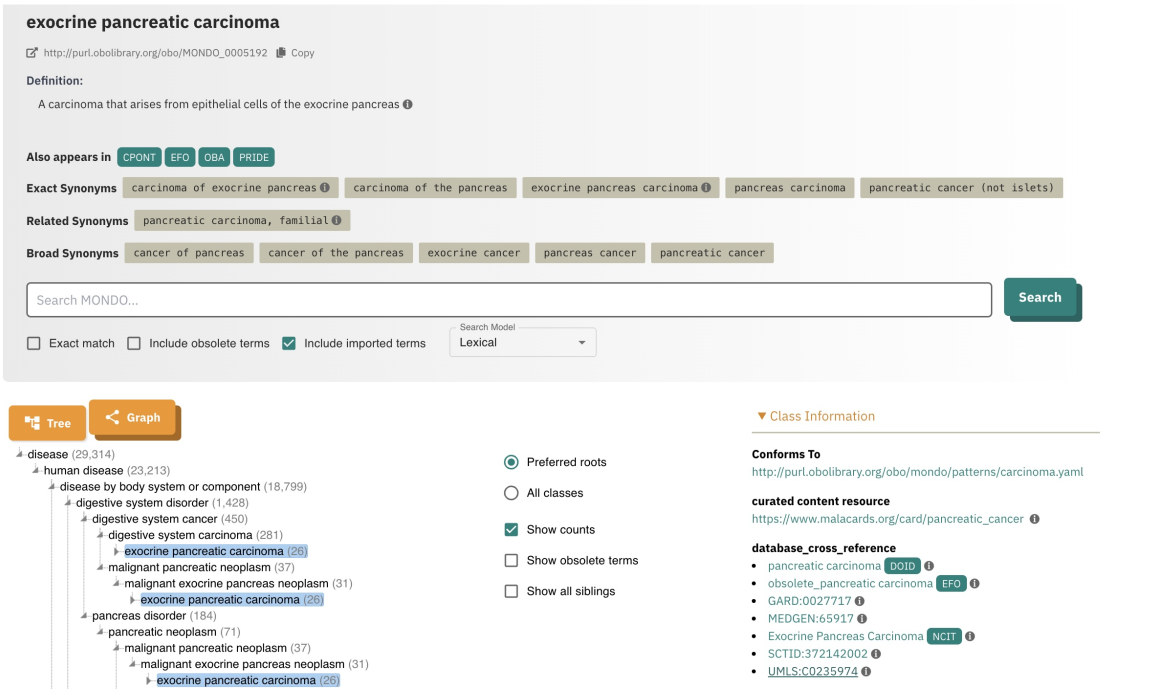


Figure 1.1: mondo snapshot

Key elements:

- definition of the short phrase or "term"
- "IRI": a URL-like entity that can be used for web access
- "CURIE": compact universal resource identifier of fixed length (MONDO:0005192)
- snapshot of conceptual hierarchy "tree"

Let's program with the MONDO ontology. We need to have the devtools and remotes packages installed, and then we can install the ontoProc2 package with `BiocManager::install("vjcitn/ontoProc2")`. Once this is done, the following will succeed.

```
library(ontoProc2)
## OpenTelemetry error: there is no package called 'otelsdk'
## OpenTelemetry error: there is no package called 'otelsdk'
mondo = semsql_connect(ontology="mondo") # download on first attempt
## Connected to SemanticSQL database: /root/.cache/R/BiocFileCache/1997777ad18_mondo.db
## Primary ontology prefix: MONDO
report(mondo)
##
## ============================================================
```

```
## SemsqlConn Object
## ============================================================
##
## Connection Details:
## ----------------------------------------
##   Database path:    /root/.cache/R/BiocFileCache/1997777ad18_mondo.db
##   Ontology prefix:  MONDO
##   Status:              Connected
##
## Database Statistics:
## ----------------------------------------
##   Labeled terms:    59,064
##   Direct edges:     109,103
##   Entailed edges:   5,003,896
##   Definitions:      22,594
##
## Terms by Prefix (top 5):
## ----------------------------------------
##   MONDO:           31,886
##   UBERON:          5,359
##   HGNC:            5,023
##   HP:              4,837
##   GO:              4,072
##
## Key Tables Available:
## ----------------------------------------
##     rdfs_label_statement
##     has_text_definition_statement
##     edge
##     entailed_edge
##     rdfs_subclass_of_statement
##     owl_some_values_from
##     has_oio_synonym_statement
##
## ============================================================
## Use methods like search_labels(), get_ancestors(), etc.
## Run ?SemsqlConn for documentation.
## ============================================================
```

To get precise about phenotypes of interest, we generally want to be able to manipulate the associated CURIEs. To learn them for a word or phrase of interest, `search_labels` can be used. We'll focus on "asthma" for now.

```
al = search_labels(mondo, "asthma")
al
##          subject                                          label
## 1    CHEBI:49167                           anti-asthmatic drug
## 2    CHEBI:65023                          anti-asthmatic agent
## 3   ECTO:9001829               exposure to anti-asthmatic drug
## 4     HP:0002099                                        Asthma
## 5   MAXO:0000634                  anti-asthmatic agent therapy
## 6  MONDO:0001491                          cough variant asthma
## 7  MONDO:0004765                              intrinsic asthma
## 8  MONDO:0004766                            status asthmaticus
## 9  MONDO:0004784                               allergic asthma
## 10 MONDO:0004979                                        asthma
## 11 MONDO:0005405                        childhood onset asthma
## 12 MONDO:0008834 asthma, nasal polyps, and aspirin intolerance
## 13 MONDO:0008835       asthma, short stature, and elevated IgA
## 14 MONDO:0010940            inherited susceptibility to asthma
## 15 MONDO:0011805   asthma-related traits, susceptibility to, 1
## 16 MONDO:0012067   asthma-related traits, susceptibility to, 2
## 17 MONDO:0012379   asthma-related traits, susceptibility to, 3
## 18 MONDO:0012577   asthma-related traits, susceptibility to, 4
## 19 MONDO:0012607   asthma-related traits, susceptibility to, 5
## 20 MONDO:0012666   asthma-related traits, susceptibility to, 6
```

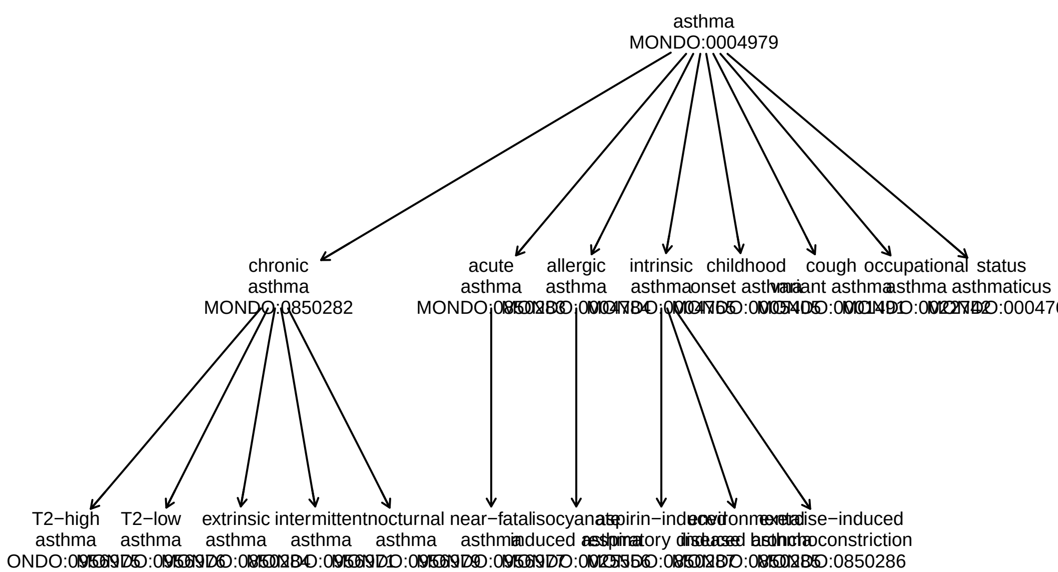
Now that we know an associated CURIE, we can get additional information about its position in the conceptual hierarchy of diseases defined in MONDO.

```
ainf = get_term_info(mondo, "MONDO:0004979")
str(ainf)
## List of 6
##  $ id          : chr "MONDO:0004979"
##  $ label       :'data.frame':   1 obs. of  2 variables:
##   ..$ subject: chr "MONDO:0004979"
##   ..$ label  : chr "asthma"
##  $ definition  : chr "A bronchial disease that is characterized by chronic inflammation
##  $ synonyms    :'data.frame':   0 obs. of  3 variables:
##   ..$ subject  : chr(0)
##   ..$ predicate: chr(0)
##   ..$ synonym  : chr(0)
##  $ superclasses:'data.frame':   1 obs. of  2 variables:
##   ..$ id   : chr "MONDO:0001358"
##   ..$ label: chr "bronchial disorder"
```

```
##   $ subclasses  :'data.frame':   8 obs. of  2 variables:
##    ..$ id   : chr [1:8] "MONDO:0850283" "MONDO:0004784" "MONDO:0005405" "MONDO:0850282" .
##    ..$ label: chr [1:8] "acute asthma" "allergic asthma" "childhood onset asthma" "chroni
tta = get_descendants(mondo, "MONDO:0004979")
str(tta)
##  'data.frame':   18 obs. of  3 variables:
##   $ id       : chr  "MONDO:0956975" "MONDO:0956976" "MONDO:0850283" "MONDO:0004784" ...
##   $ label    : chr  "T2-high asthma" "T2-low asthma" "acute asthma" "allergic asthma" ...
##   $ predicate: chr  "rdfs:subClassOf" "rdfs:subClassOf" "rdfs:subClassOf" "rdfs:subClassOf
```

We will use an alternate representation of the MONDO ontology that is useful for visualizing relationships.

```
data("monoi", package="CSHLvc2026")
onto_plot2(monoi, tta$id, cex=.55)
```



1.2 Study Designs

GWAS can be conducted under several study designs, each with distinct strengths and confounders. This section compares case-control studies, cohort studies, family-based designs, and biobank-scale analyses, with attention to population stratification and the importance of ancestry-matched controls.

1.3 Manhattan Plots

The Manhattan plot is the canonical visualization for GWAS results, displaying $-\log_{10}(p)$ values across the genome. This section explains how to interpret Manhattan plots, choose significance thresholds (genome-wide significance at $p < 5 \times 10^{-8}$), and identify credible association signals amid linkage disequilibrium patterns.

1.4 Functional Interpretation

Identifying a significant SNP is only the beginning. This section covers post-GWAS functional annotation — including regulatory element overlap, eQTL colocalization, gene prioritization strategies, and pathway enrichment — to move from statistical association to biological insight.

2 xQTL

This chapter covers expression and molecular quantitative trait loci (xQTL), connecting genetic variation to intermediate molecular phenotypes as a bridge between GWAS signals and biological mechanisms.

3 Spatial Transcriptomics

This chapter introduces spatial transcriptomics technologies and analytical frameworks, exploring how the spatial context of gene expression enriches our understanding of tissue architecture and disease.

4 Future Computing

This chapter looks ahead at emerging computational paradigms relevant to genomics and biomedical research, including cloud-native workflows, federated analysis, and the role of AI/ML in large-scale data integration.